Parametric Pavilion

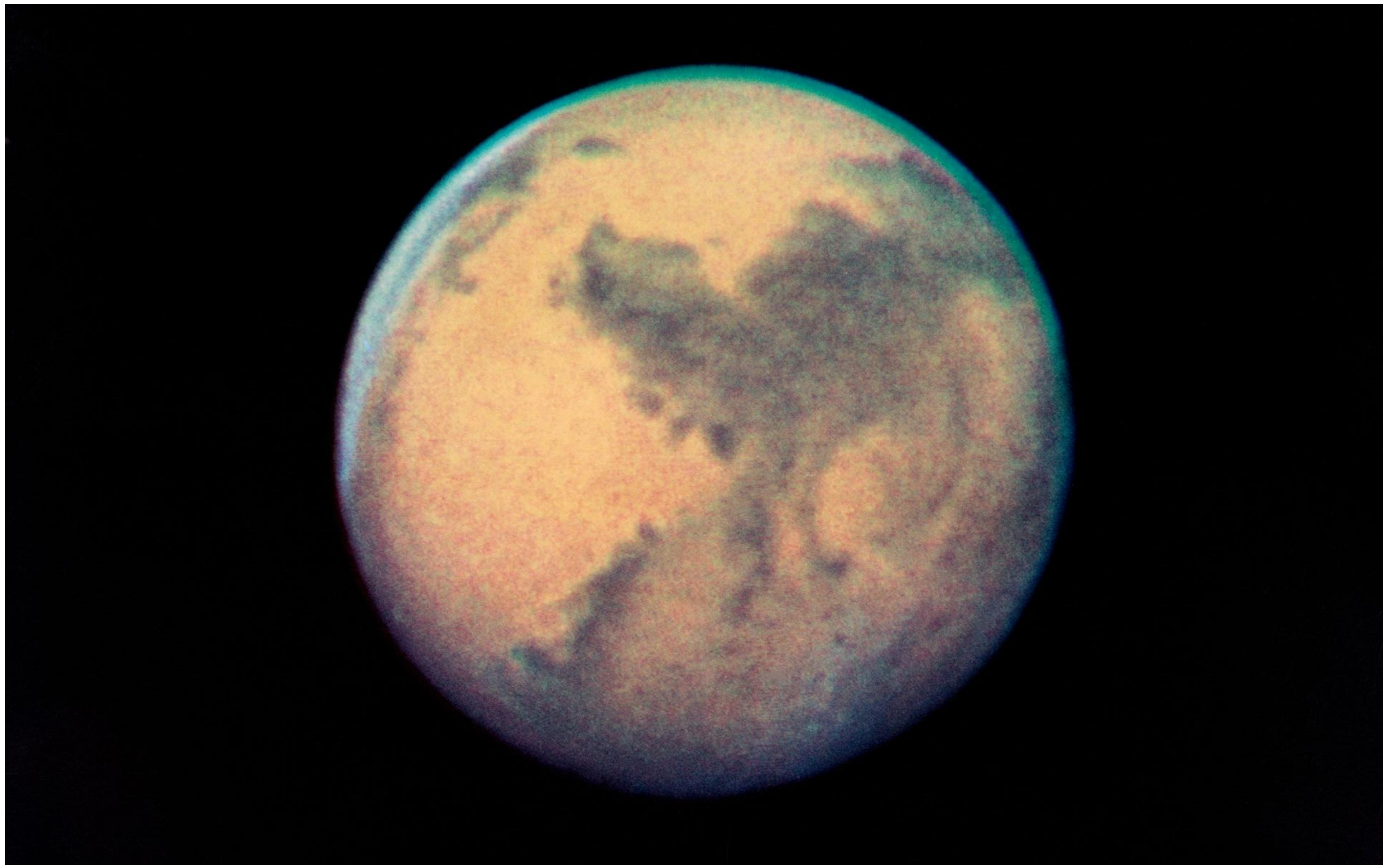


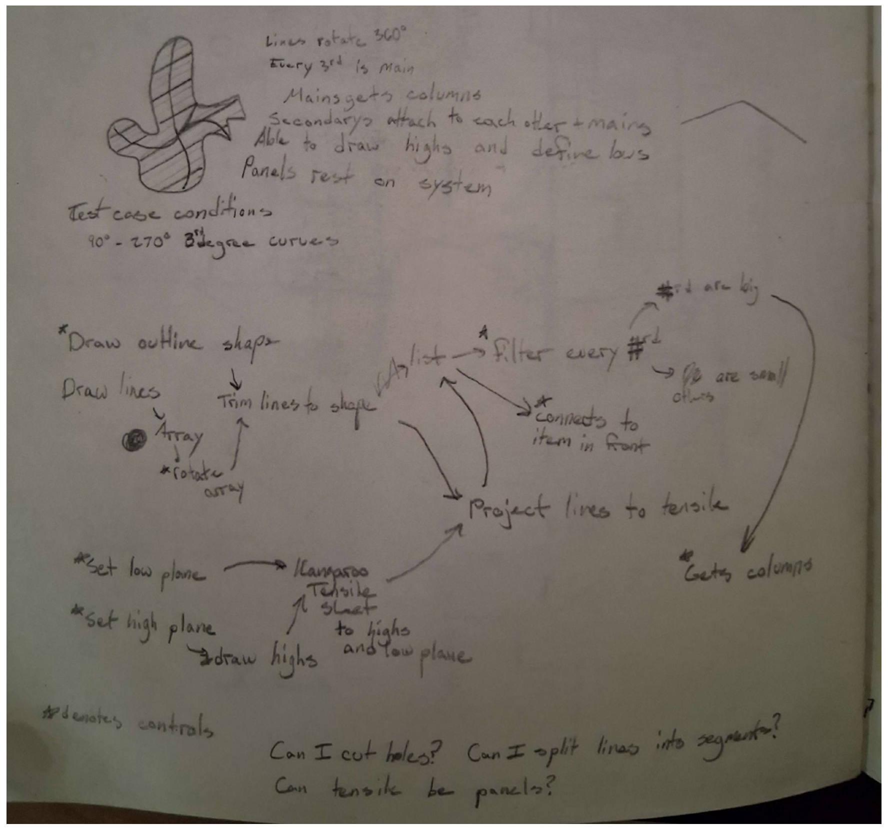
Photo: NASA at https://science.nasa.gov/image-detail/amf-s91-32389/

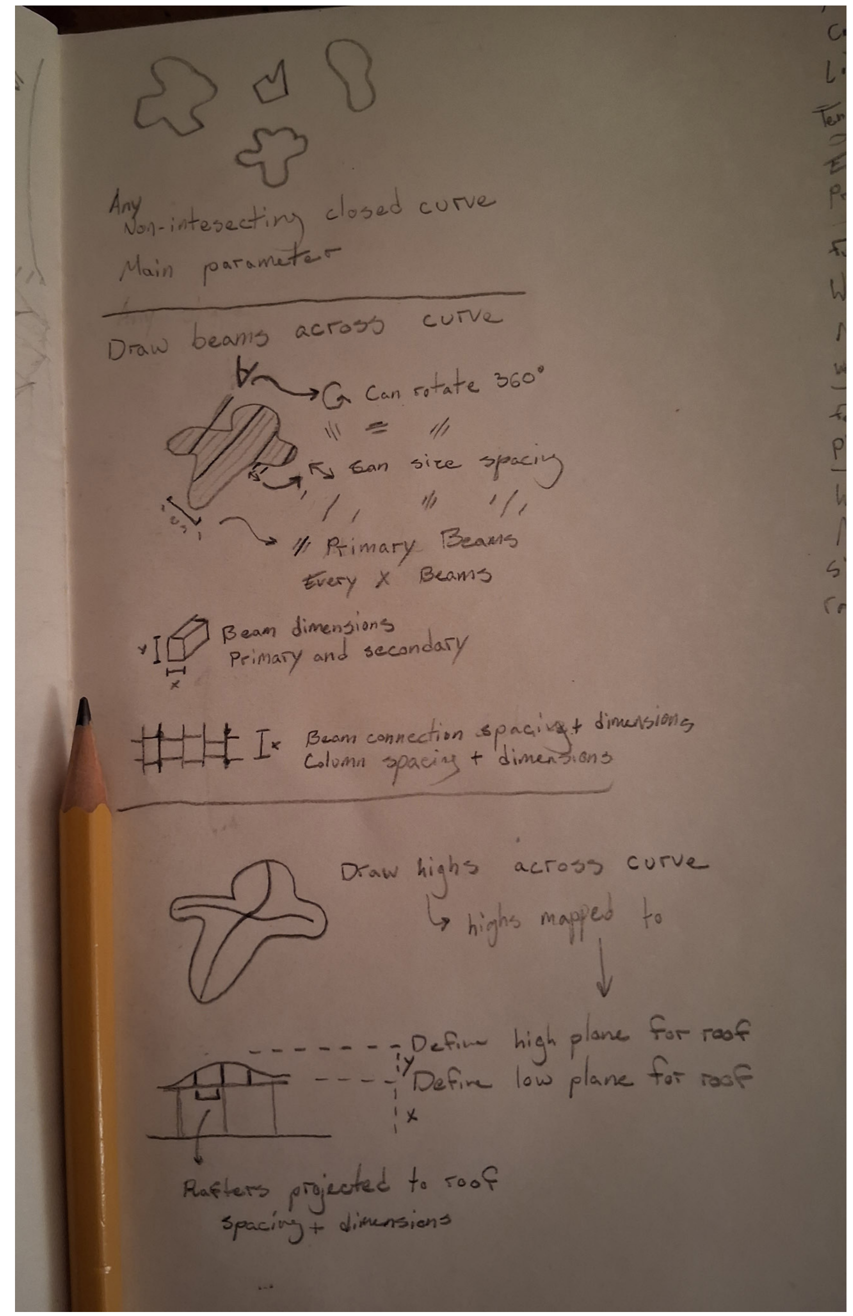
Arch565_Assignment 2

by Alex Tupper

1. Sketching

Journal Entries





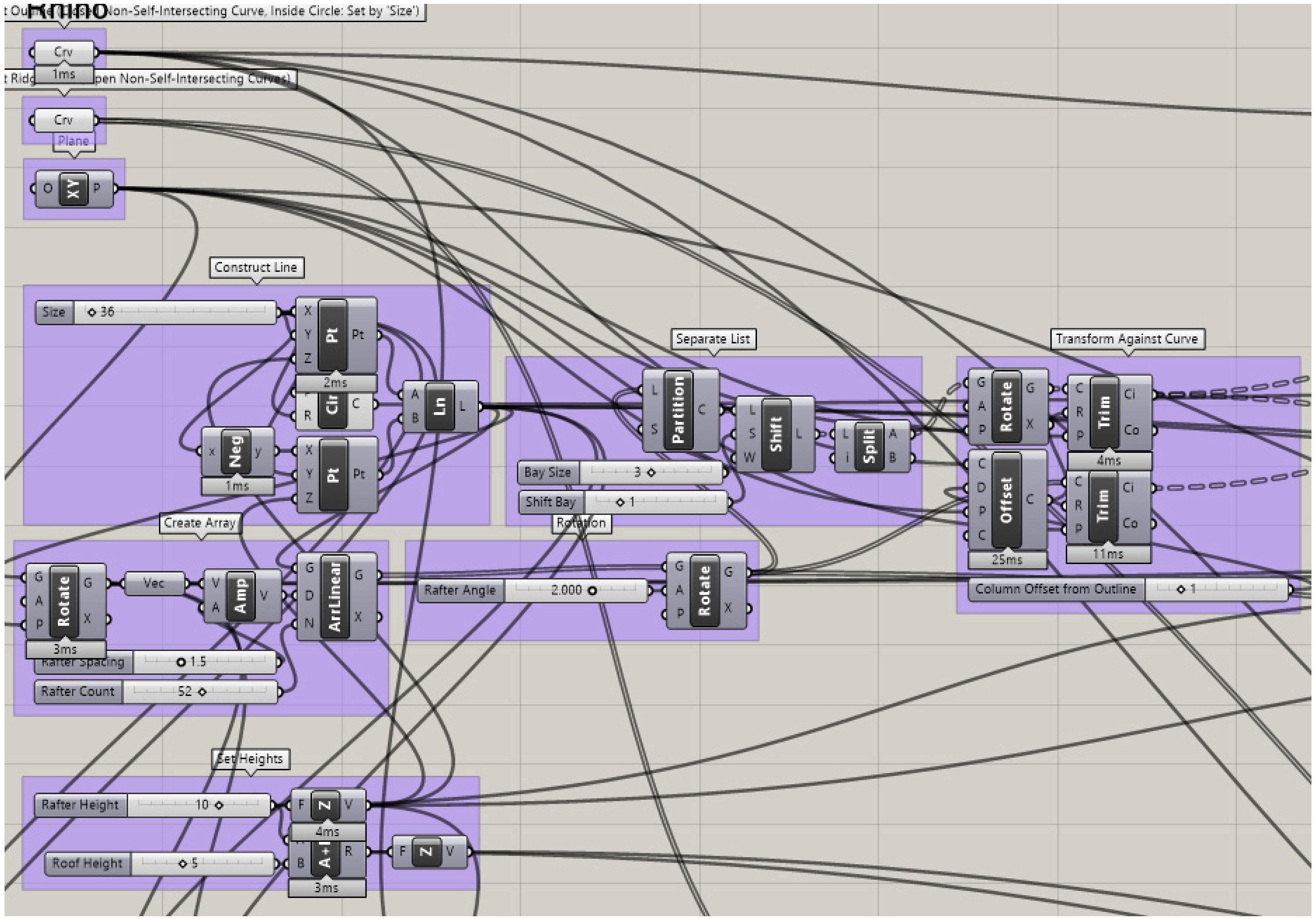
These are my initial sketchs for my pavilion. I chose to make a simple covering, but one that could be parametrically modeled with as few inputs as possible.

Last semester, I tried and failed to make a covered walkway because I did not have the technical ability to model it in Revit. This semester, with the power of Rhino Inside Revit, I wanted to go back and try again.

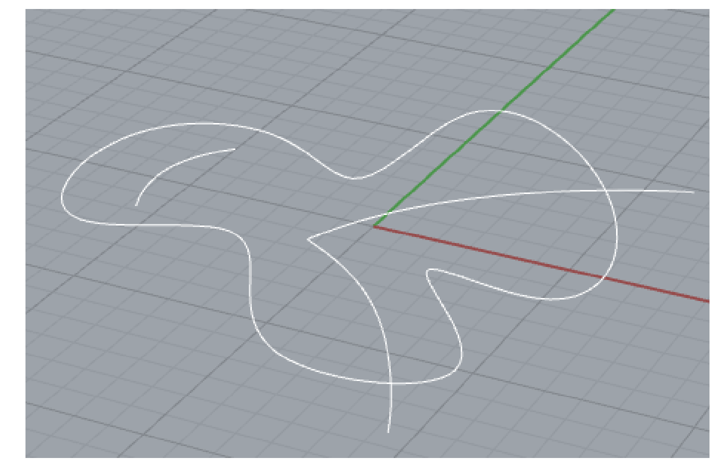
I am glad I did this because I came up with a result that I am happy with. I follwed my intial ideas pretty closley, although there were a bunch of things I had to get more detailed with. The roof was the hardest and most frustrating part because so many of my iterations did not do what I wanted with as few inputs as I was willing to give it. In the end I figured it out with a lot of hard work and I am extremely happy with the outcome.

2. Grasshopper Script

Top Left



This is the top left of the script. The Crv components take the initial outline curve and the curves for the ridge line of the roof. These are the only input geometries that need to be drawn in Rhino.



Example Input Geometries

On the bottom of this image is the rafter and roof heights.

The rest of the groups set up the array of lines that will become the beams and joists with variables to control rotation and spacing.

2. Grasshopper Script

Bottom Banner

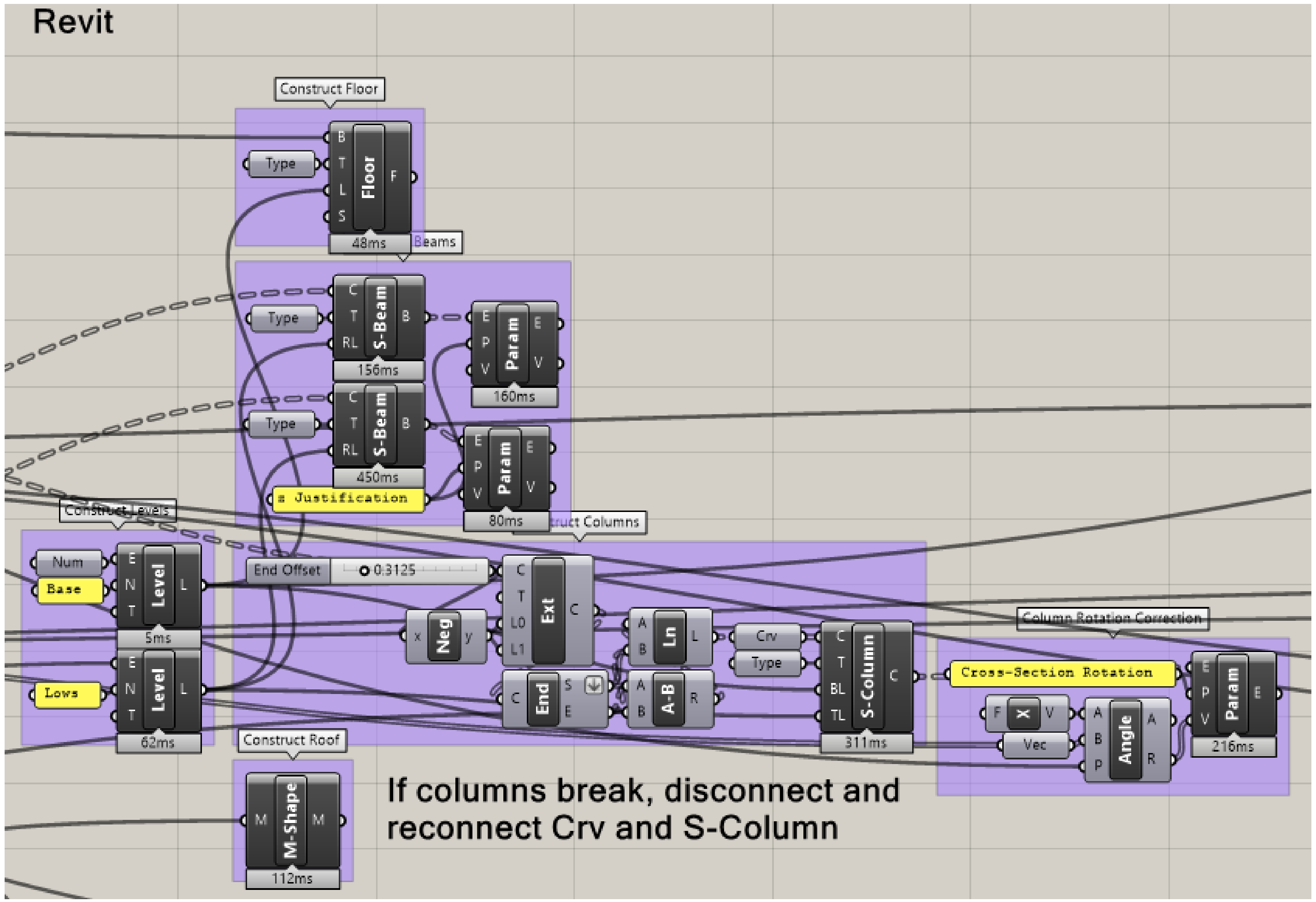
This bottom section generates the roof. This is the part that took the longest, mostly because I went through five ideas on how to implement it. I started with Kangaroo and tried a couple of methods with that before I switched gears. I tried generating meshes with drape commands, but that also failed.

Finally I decided to construct a set of points that I could move up and down, and turn that into a mesh. When I got the points to move how I wanted them to I literally jumped out of my chair in happiness. There was a lot of googling, as seen by the attributions, but making the points move was all me, which feels pretty good.

The gist of what is happening is that a planar mesh is deconstructed to points, which are then moved vertically corresponding with their distance to the ridge lines. These points are turned back into a mesh which then gets cleaned up and extruded into a 3d mesh.

3. Rhino Inside Revit

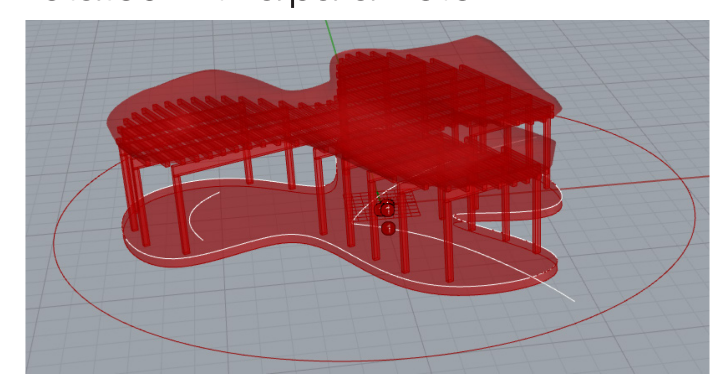
Mid Section



This portion of the script takes various geometries and turns them into revit components.

Levels are constructed at 0 and on the rafter height.

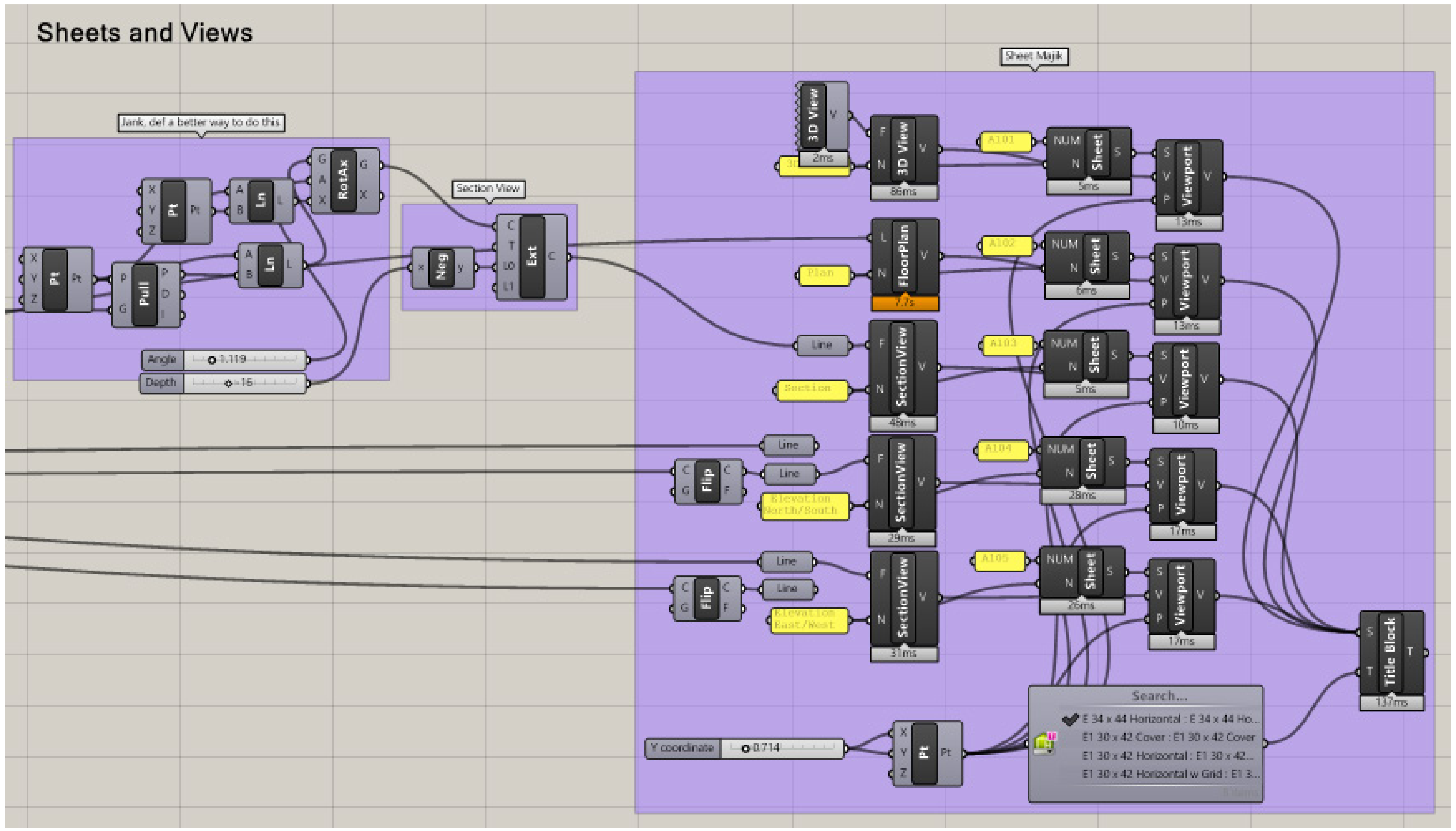
The columns are constructed here, although this could have been grouped with the rest of the Rhino components. For some reason the columns break sometimes, but disconnecting and reconccting it fixes them. They also had to be rotated with a parameter.



Example Components Seen in Rhino

3. Rhino Inside Revit

Top Right

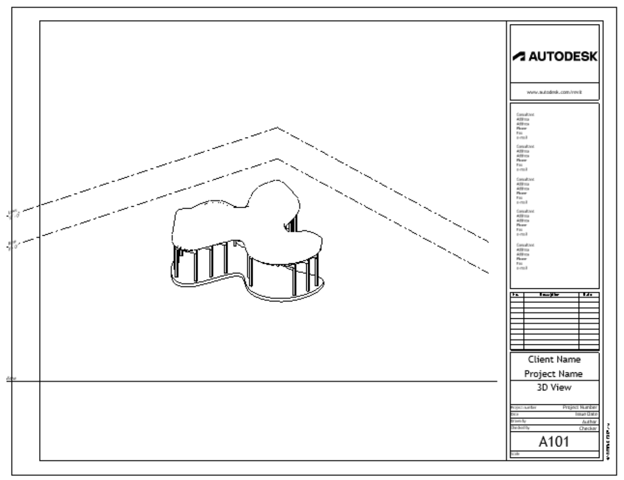


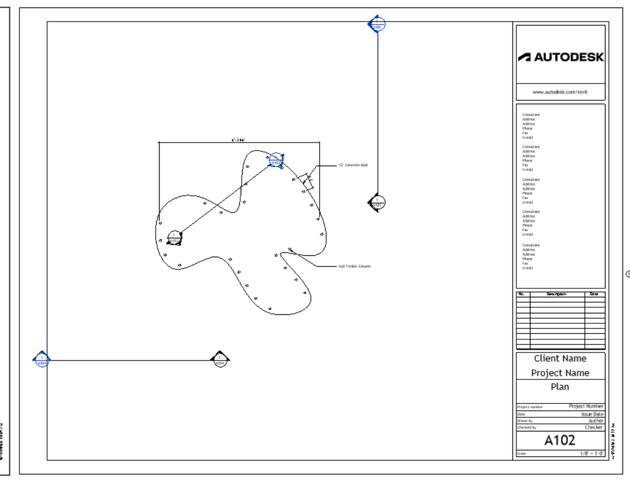
This is the portion that creates all the views, viewports, and sheets used by Revit.

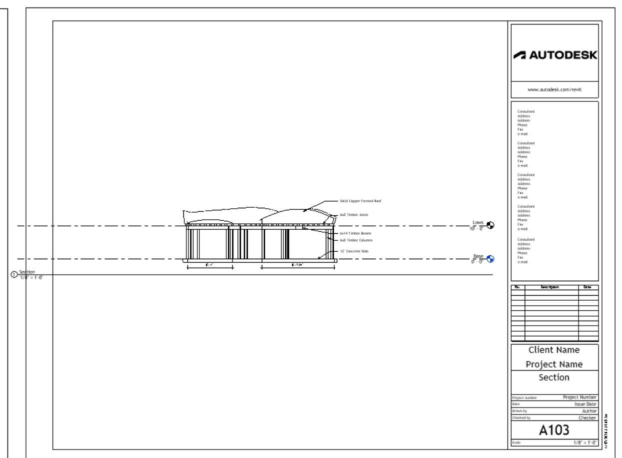
It wasn't too hard to figure out, but its kind of tedious to build this stuff in Grasshopper. Not sure what the use case is for doing this here instead of in Revit.

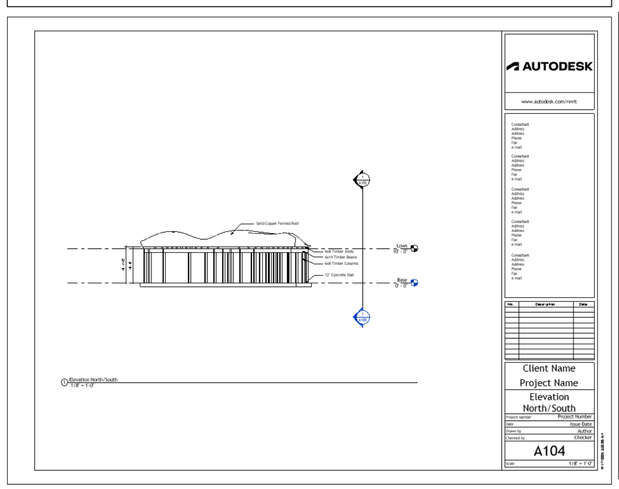
3. Rhino Inside Revit

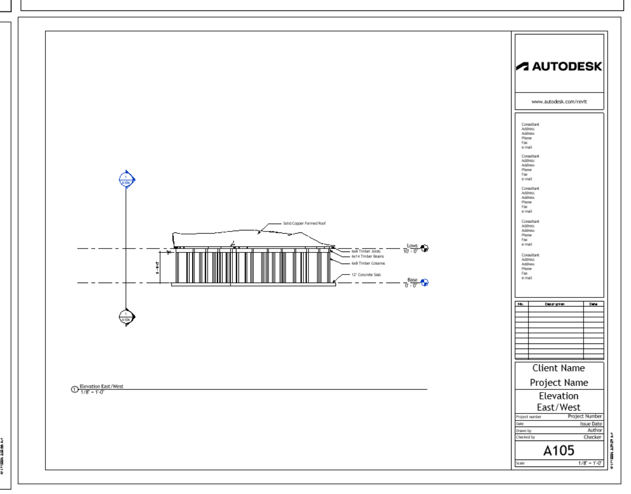
Revit







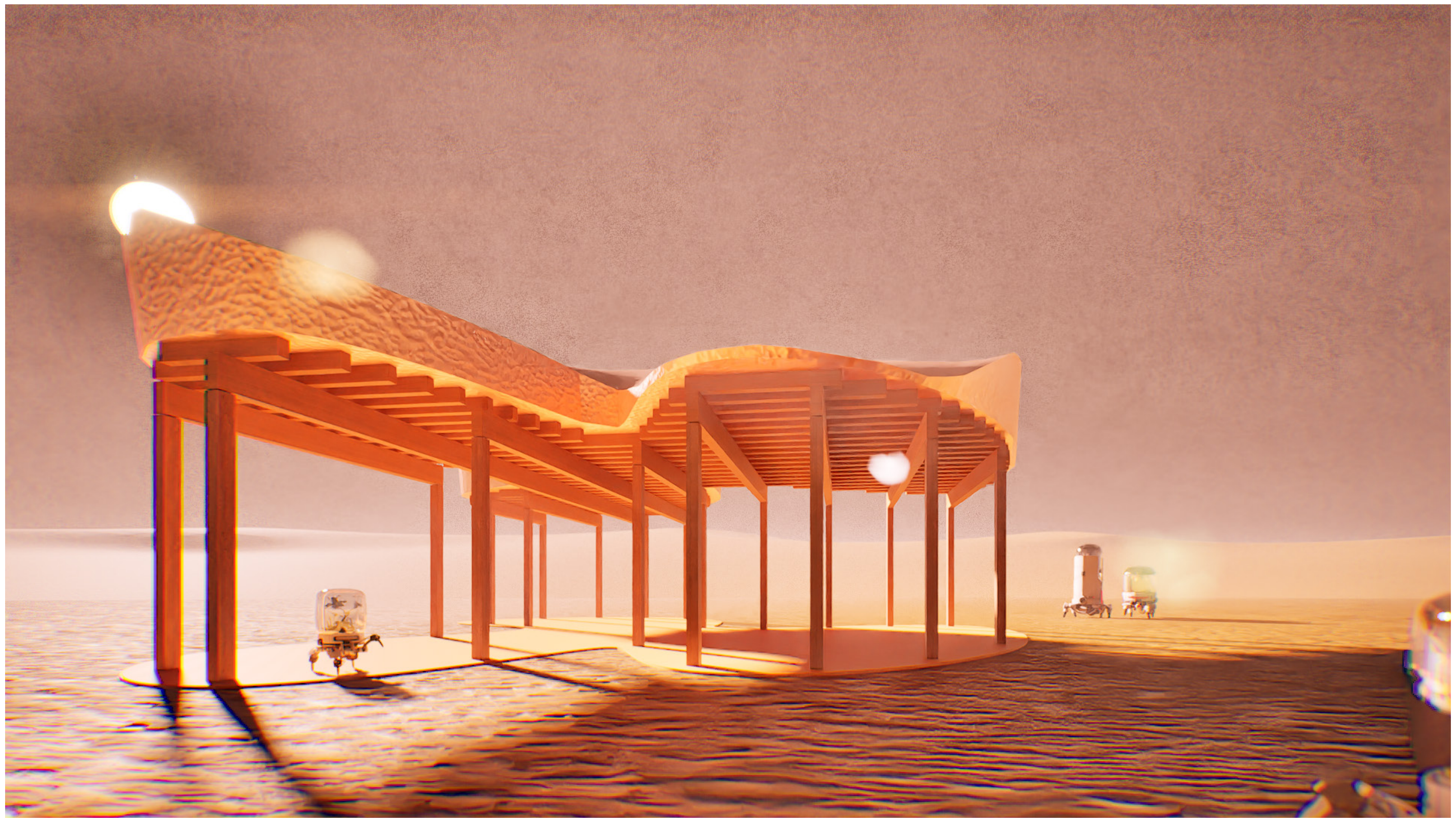




As seen by the dimension lines, this pavilion isn't construction accurate, but I am still quite happy with the way the script turned out.

4. Revit Inside Twinmotion

Renders



I have used Twinmotion before, so in all honesty I did not learn much from this portion of the class. I decided to have fun with my renders and put my pavilion on Mars.

CC attributions to users N01516 and ChrisLee for uploading the robot models to the Unity assest store. I get the feeling however that these were stolen from an uncredited artist.

4. Revit Inside Twinmotion

Renders Cont.